



Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
OPTICAL AND ELECTRICAL
MEASUREMENTS OF AC-LED
PACKAGES AND ARRAYS OR MODULES
AN AMERICAN NATIONAL STANDARD



ANSI/IES LM-88-18(R2023)

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has been approved by IES.
Suggestions for revisions
should be directed to IES.



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CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Scope	1
2.0	Normative References	1
2.1	IES LM-85-14	1
2.2	IES RP-16-17	1
3.0	Definitions	2
3.1	AC-LED	2
3.2	AC-LED package	2
3.3	AC-LED array or module	2
3.4	device under test (DUT)	2
3.5	remote-phosphor AC-LED package	2
3.6	remote-ohosphor AC-LED array or module	2
3.7	source measure unit (SMU)	2
3.8	temperature-controlled mount (TCM)	2
4.0	Optical Measurement Instruments	2
5.0	Electrical Measurement Instruments	2
5.1	Performance of Electrical Measurement Instruments	2
5.2	Measurement Uncertainties of Electrical Measurement Instruments	2
6.0	Preparation for Measurements	3
6.1	Seasoning	3
6.2	Operating Orientation	3
6.3	Thermal Conditions	3
6.3.1	Junction Temperature	3
6.3.2	Ambient Air Temperature	3
6.4	Airflow	3
6.5	Humidity	3

7.0	Method of Measurement under Single-Cycle AC Operation	3
7.1	General	3
7.2	Measurement Procedure	4
7.3	Estimation of the Measurement Uncertainty	4
8.0	Method of Measurement under Continuous AC Operation	6
8.1	General	6
8.2	Measurement Procedure	6
8.3	Methods Used to Determine the Initial Reference Current, $I_F(0)$	8
8.3.1	RMS Initial Reference Current, $I_F(0)_{rms}$	8
8.3.2	Instantaneous Initial Reference Current, $I_F(0)_{ins}$	8
8.4	Estimation of the Measurement Uncertainty Due to Junction Temperature Rise	9
9.0	Optical Measurements	10
10.0	Test Report	10
10.1	General Information	10
10.2	Description of DUT	10
10.3	Description of Measurement System	10
10.4	Environmental Conditions	10
10.5	Measurement Conditions	10
10.6	Description of Measurement	10
10.7	Optical Quantities (Time-Averaged) (Typical)	11
10.8	Electrical Quantities (Typical)	11
Additional Reading		11
Informative References		11

1.0 Introduction and Scope

1.1 Introduction

This document describes methods of measurements for alternating-current-driven light emitting diodes (AC-LEDs), including AC-LED packages and AC-LED arrays or modules (see **Section 3** for definitions) that are designed to operate on an AC source such as a 120-V, 60-Hz main without the need of a driver. This document provides uniform methods for operation and measurement of AC-LEDs. It is important to note that the methods and instruments for optical measurements are not described in this document; for such information, the reader is referred to IES LM-85-14 (see **Section 2.1**).

Lighting practitioners have found it difficult to obtain accurate optical and electrical characteristics of AC-LEDs at realistic elevated-temperature operating conditions. It is common practice for AC-LED manufacturers to use single-cycle AC operation at 25 °C with no heat sink for optical and electrical measurements. This assumes that the junction temperature of an AC-LED under these conditions is equal to its ambient air temperature (typically 25 °C); therefore, published specifications of AC-LEDs normally refer to the junction temperature of 25 °C. AC-LEDs are, however, typically operated at a much higher junction temperature (varying from 50 °C to more than 100 °C). Under such elevated thermal conditions, the optical and electrical characteristics of an AC-LED tend to deviate significantly from those of operation at a junction temperature of 25 °C, as a result of their strong dependence on the AC-LED's thermal condition. To address this issue, some AC-LED manufacturers provide data of optical and electrical characteristics at elevated operating temperatures; however, such data are not always available. In addition, there have been no standard test methods published for measuring AC-LEDs operated at an elevated temperature.

The test methods described in this document are based on setting an AC-LED of any type to a specified junction temperature during optical and electrical measurements under either single-cycle AC operation (**Section 7**) or continuous AC operation (**Section 8**). The single-cycle AC operation method is fast and is commonly used by AC-LED manufacturers for rapid measurements at the

junction temperature of 25 °C, while the continuous AC operation method is slow but is often used by end users and testing laboratories to achieve small measurement uncertainties.

In cases where the intent is to determine the characteristics of an AC-LED when operated under something other than the standard conditions described in this document, the measured results are valid only for the particular condition. Any deviation from the standard conditions shall be clearly described in the test report.

1.2 Scope

This approved method describes the procedures to be followed and precautions to be observed in performing optical and electrical measurements of AC-LEDs for total luminous flux, total radiant flux, total photon flux, color characteristics, electrical voltage, current, and power. This approved method applies to AC-LED packages, remote-phosphor AC-LED packages, AC-LED arrays or modules, and remote-phosphor AC-LED arrays or modules. This document specifies measurement methods under single-cycle AC operation as well as continuous AC operation with an AC frequency of 50 Hz or 60 Hz; in both cases, "thermal conditions" of AC-LEDs refers to their junction temperatures.

This document does not cover LED lamps, LED light engines, or LED luminaires; it also does not cover DC-LEDs, which are within the scope of IES LM-85-14 (**Section 2.1**).

2.0 Normative References

2.1 IES LM-85-14

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of High-Power LEDs. New York: The Organization; 2014 (or the latest version).

2.2 IES RP-16-17

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: The Organization; 2017.

3.0 Definitions

3.1 AC-LED

An LED with multiple p-n junctions that is designed to operate on AC sources.

3.2 AC-LED package

An LED package that is designed to operate on AC sources.

3.3 AC-LED array or module

An LED array or module that is designed to operate on AC sources.

3.4 device under test (DUT)

The AC-LED package, remote-phosphor AC-LED package, AC-LED array or module, or remote-phosphor AC-LED array or module that is being tested.

3.5 remote-phosphor AC-LED package

An AC-LED package containing a phosphor that is thermally separated from the pump or source radiation element but contained within the AC-LED package.

3.6 remote-phosphor AC-LED array or module

An AC-LED array or module containing a phosphor that is thermally separated from the pump or source radiation element but contained within the AC-LED module or array.

3.7 source measure unit (SMU)

An electrical measurement instrument that can source voltage or current and measure current or voltage (or both current and voltage simultaneously) with a source-delay-measure sequence.

3.8 temperature-controlled mount (TCM)

A device that incorporates cooling and heating components (typically thermoelectric coolers and/or heaters) to maintain its temperature at a set value.

4.0 Optical Measurement Instruments

The instruments used for optical measurements are described in IES LM-85-14 (see **Section 2.1**). If the single-cycle AC method is used (see **Section 7**), the photometer or the spectroradiometer shall be capable of measuring a single optical pulse with the desired spectral range.

5.0 Electrical Measurement Instruments

5.1 Performance of Electrical Measurement Instruments

The sampling speed of a multimeter used for the single-cycle AC operation method (**Section 7**) or continuous AC operation method (**Section 8.3.1** and **Section 8.3.2.1**) shall be 20,000 readings per second or higher. The speed of an SMU (see **Section 3.7**) used for the continuous AC operation method (**Section 8.3.2.2**) shall be 10,000 samples per second or higher. The AC power supply shall not have a voltage total harmonic distortion greater than 3%.

5.2 Measurement Uncertainties of Electrical Measurement Instruments

The measurement uncertainty of an electrical measurement instrument depends on the selected range and the measurement speed. The instruments used for the electrical measurements of devices under test (DUTs) shall be capable of making root mean square (RMS) measurements with these uncertainties:

- AC voltage and current: no greater than 0.2%
- AC power: no greater than 0.5%.

Note 1: Here, uncertainty refers to relative expanded uncertainty with a 95% confidence interval, normally with a coverage factor of $k=2$, as prescribed by the standards.^{1,2}

6.0 Preparation for Measurements

6.1 Seasoning

Seasoning is not required for the methods described in this document; however, if a device under test (DUT) is seasoned, all operating conditions and durations shall be reported.

6.2 Operating Orientation

The light emission process of a DUT is not affected by its orientation. The effect of variation of thermal convection due to different operating orientation on the measured optical and electrical characteristics of a DUT is typically small with the methods described in this document, with the exception of remote-phosphor type DUTs. The phosphor temperature of a remote-phosphor type DUT may vary with the operating orientation, resulting in the variation of the DUT's optical and electrical characteristics. Therefore, the operating orientation of a remote-phosphor type DUT shall be reported.

6.3 Thermal Conditions

6.3.1 Junction Temperature. Optical and electrical characteristics of a DUT vary with its junction temperature, T_j , which is the average temperature of all p-n junctions in this case. The junction temperature of a DUT, T_j , can be set by the methods provided in **Section 7** and **Section 8**. Reports of optical and electrical values of a DUT shall include the junction temperature used in the measurement, which is in fact the average junction temperature over one AC cycle. Junction temperature T_j at 25 °C is typically chosen as the nominal condition; however, elevated junction temperatures—e.g., 50 °C, 100 °C—are also used to represent the temperature of the DUT operating in lighting products.

6.3.2 Ambient Air Temperature. Ambient air temperature, T_a , is critical if it is used to set the junction temperature for the single-cycle AC measurement (see **Section 7**). In such a case, the ambient air temperature shall be measured at a point not more than 0.5 m from the DUT and at the same height as the DUT, using a temperature sensor shielded from direct radiation from the DUT. If an integrating sphere is used for the optical measurement, *ambient air temperature* refers

to the temperature inside the sphere, which should be measured under the same condition as described above, or in an alternative condition if this is not possible due to the design and/or the size of the integrating sphere.

If the junction temperature is controlled by a temperature-controlled mount (TCM), the DUT's optical and electrical characteristics are normally not sensitive to its ambient air temperature, with the exception of remote-phosphor type DUTs, for which the ambient air temperature, T_a , shall be maintained at $T_j \pm 5$ °C and reported.

6.4 Airflow

Ambient airflow may affect optical and electrical characteristics of a DUT and should be minimized. (Note: Optical and electrical characteristics of a DUT are normally not sensitive to the ambient airflow for the methods described in this document, with the exception of remote-phosphor type DUTs.) The phosphor temperature of a remote-phosphor type DUT may vary with the ambient airflow speed, resulting in the variation of the DUT's optical and electrical characteristics. Thus, for measurement of a remote-phosphor type DUT, the ambient airflow speed should be reported.

6.5 Humidity

Humidity may affect optical and electrical characteristics of a DUT, depending on the type of DUT. It should be maintained to less than 65% relative humidity, and the actual humidity should be reported.

7.0 Method of Measurement under Single-Cycle AC Operation

7.1 General

This method can be used for measurement of any type of device under test (DUT). When this single-cycle AC operation method is used, a DUT is operated and measured for a period of a single 50-Hz or 60-Hz AC power cycle. Under this operation, the junction temperature of the DUT is assumed to be approximately the same as its ambient air temperature or its temperature-controlled mount (TCM) temperature.

This method uses two fast multimeters for synchronized measurements of the instantaneous voltage and current, an AC power supply, and optical measurement instruments that are able to measure pulsed optical radiation from the DUT.

7.2 Measurement Procedure

The following measurement steps shall be taken.

Step 1: Stabilizing the DUT to the specified junction temperature.

For the nominal condition of T_j at 25 °C, the junction temperature of a DUT is normally controlled by the laboratory's ambient air temperature of $T_a = 25$ °C. For an elevated junction temperature condition, a TCM is normally used.

When using the ambient air temperature control, the ambient air temperature is set to the desired DUT junction temperature. When using a TCM, the DUT is mounted on the TCM and the temperature of the TCM (T_{TCM}) is adjusted to achieve the specified junction temperature.

Note 2: *The TCM temperature (T_{TCM}) can be set to the same as the specified junction temperature in most cases because the temperature difference between the junction and the TCM is negligible after the system reaches thermal equilibrium. Also, when the estimated rise of junction temperature is known (see **Section 7.3**) the ambient air temperature (T_a) or TCM temperature (T_{TCM}) should be adjusted lower so that the actual junction temperature during the measurement is the same as that specified; thus, the measurement errors resulting from the junction temperature rise are corrected.*

In both cases, the DUT shall be thermally stabilized. The thermal stability of the DUT shall be monitored by measuring the *changing* average optical output or the *changing* RMS current of a single-cycle AC pulse using the same method as that used in Step 2, or by measuring the temperature at its thermal contact point (pin, case, or board). The DUT temperature is considered stabilized if at least 10 consecutive average optical output measurements, or at least 10 consecutive RMS current measurements of single-cycle AC pulses

(with frequency less than 1 pulse per second), remain within 0.3% for at least 1 minute, or if the measured temperature at its thermal contact point remains within ± 1 °C of the specified junction temperature for at least 1 minute.

Step 2: Performing optical and electrical measurements.

The specified AC power is applied to the DUT, starting at the AC phase angle of 0 degrees, using a phase angle controllable AC power supply. Optical and electrical measurements are started at the same time, at 0 degrees phase angle. The measurements are stopped at the phase angle of 360 degrees, and the AC power is turned off (see **Figure 1**). Due to the dead time near the phase angles 0 degrees and 360 degrees, accuracy of the synchronization between applying AC voltage at phase angle 0 degrees and starting optical or electrical measurements is not critical.

Within one AC power cycle, an individual p-n junction of the DUT may be activated in the first half of the AC cycle, the second half of the AC cycle, or in both the first and second halves of the AC cycle, depending on the design of the DUT; therefore, the optical and electrical measurement time shall cover an entire cycle from phase angle 0 degrees. The integration time is approximately 16.7 milliseconds for AC power of 60 Hz (shown in **Figure 1**), and 20 milliseconds for AC power of 50 Hz. The exact period or frequency of the AC power shall be measured, recorded, and used for converting the measured single-cycle AC operation optical and electrical quantities to those under continuous AC operation.

7.3 Estimation of the Measurement Uncertainty

The junction temperature of a DUT rises rapidly during the first AC cycle (60 Hz or 50 Hz). Therefore, the actual junction temperature may be considerably higher than the assumed temperature, i.e., the ambient air temperature (T_a) or TCM temperature (T_{TCM}). Furthermore, in the case of a phosphor-converted white DUT, the phosphor temperature is close to the junction temperature under this single-cycle AC operation, which is significantly lower; e.g., tens of degrees Celsius lower, than the actual phosphor temperature when the DUT is under a continuous AC operation in a lighting product.

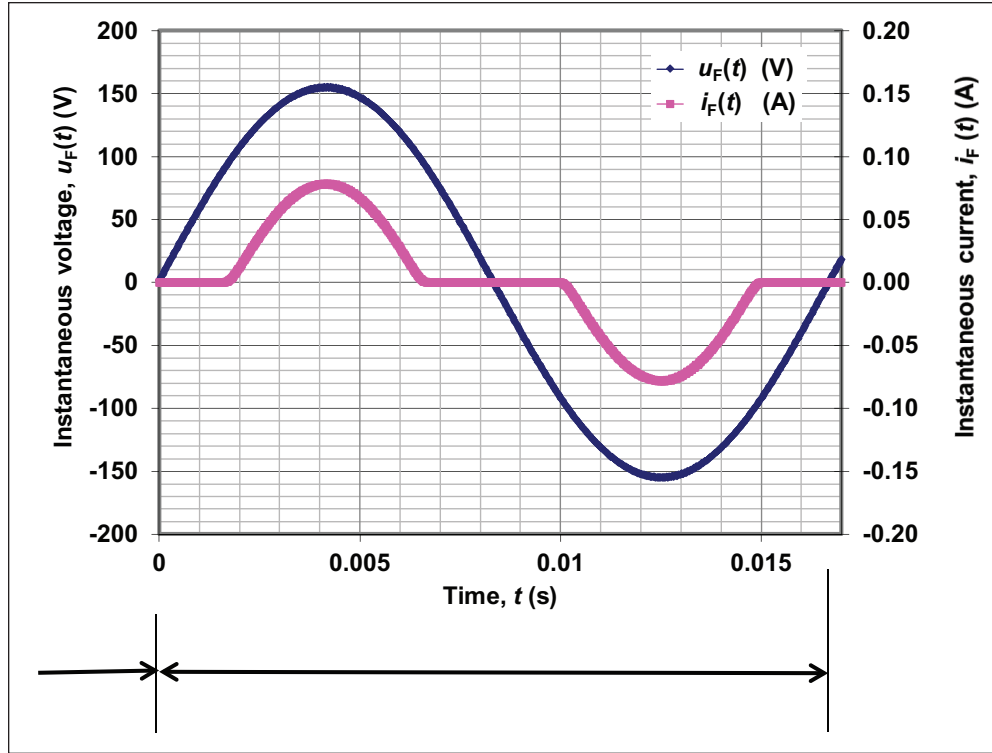


Figure 1. Timing of the single-cycle AC operation.

This rise in junction temperature should be estimated using a proper thermal model (e.g., using the model described in **Informative Reference 8**) if all design details and parameters of the DUT are known or can be measured using a proper method (e.g., using the method described in **Informative Reference 6**). Using the estimated rise of the junction temperature, the errors resulting from the junction temperature rise can be corrected during the measurement time (see **Note 2** in **Section 7.2**, Step 1). Otherwise, using the estimated rise of junction temperature, T_j , and the sensitivity coefficients of the junction temperature of the DUT for optical and electrical quantities, the measurement errors due to the rise of T_j can be estimated. Similarly, the difference between the single-cycle-AC-operation phosphor temperature and continuous-AC-operation phosphor temperature should also be estimated using a proper thermal model,^{3,4} or be measured if the phosphor is on the top of the DUT using a non-contact temperature measurement instrument, e.g., a thermal imager. Using the estimated difference of the phosphor temperature, and the sensitivity coefficients of optical and electrical quantities at the phosphor

temperature of the DUT, the measurement errors due to the difference of the two phosphor temperatures can also be estimated; however, such modeling or measurement work described above is difficult.

Alternatively, the combined measurement error due to the rise of the junction temperature and the difference of the phosphor temperatures can be assessed experimentally by comparing the measurement result with that obtained using a method that has no, or negligible, errors resulting from the rise of junction temperature during the first AC cycle. See **Section 8.2** and **Section 8.3.2**.

Corrections should be applied in order to reduce measurement uncertainties. When a correction is applied, the correction method and correction factor shall be reported, and the estimated residual error should be listed as an uncertainty component contributing to the overall measurement uncertainty. If the correction is not applied, this measurement error should be considered as a component of the overall measurement uncertainty.

8.0 Method of Measurement under Continuous AC Operation

8.1 General

When the AC current of a device under test (DUT) is accessible, the DUT can also be measured under continuous AC operation, which is the same way it is operated normally in a lighting product; therefore, such measurement error associated with the single-cycle AC operation due to the phosphor temperature difference is completely eliminated (See **Section 7.3**). This method uses a temperature-controlled mount (TCM) for accurately setting a DUT to a constant junction temperature, T_j , and controlling its feedback; therefore, optical and electrical measurements can be performed under normal AC operation, and all the conventional optical and electrical measurement instruments can be used, such as sphere-photometers, sphere-spectroradiometers, goniophotometers, gonio-colorimeters, and gonio-spectroradiometers, that are designed for measurement of traditional lamps in the lighting industry. Using this continuous-AC method, fast measurements are required only for instantaneous current and voltage, to set and control the junction temperature of the DUT. Additional information on this method is available.^{5,6}

Note 3: When a DUT is turned on, before the heat generated in its p-n junctions affects the light emission (typically a period of a few microseconds), each individual junction has the same temperature. In contrast, under a normal operating condition, the temperature of each individual junction of the DUT may vary, depending on the location of the p-n junction and its thermal resistance to the TCM; however, this variation of the junction temperature has negligible influence on the optical properties of the DUT if the average junction temperature (which can be monitored by measuring the forward current of all the DUT p-n junctions) is controlled to be the same as the initial junction temperature upon start up. This is because these p-n junctions have similar optical and electrical properties, and because the relationship between the change of optical or electrical properties and the change of the junction temperature is within the linear range; therefore, the changes of optical and electrical properties resulting from the variation of individual p-n junction temperatures are mostly cancelled.

8.2 Measurement Procedure

The following steps shall be taken to measure a DUT under the continuous AC operation:

Step 1: Mounting the DUT.

The DUT is mounted on a TCM using a metal-core printed circuit board or other mounting technique.

Step 2: Setting the junction temperature (T_j) of the DUT.

The temperature of the TCM (T_{TCM}) is set to the specified junction temperature of the DUT, and the DUT is stabilized at this temperature. The thermal stability of the DUT shall be monitored by measuring the changing initial reference current, $I_F(0)$, using the same method as that used in Step 3, or by measuring the temperature at its thermal contact point (pin, case, or board). The DUT is considered stabilized if at least 10 consecutive initial reference current measurements (with measurement frequency less than 1 measurement per second) remain within 0.3% for at least 1 minute, or if the monitored temperature remains within $\pm 1^\circ\text{C}$ of the specified junction temperature for at least 1 minute.

When the RMS initial reference current is used (see **Section 8.3.1**), and the estimated difference of junction temperature is known (see **Section 8.4**), the ambient air temperature (T_a) or TCM temperature (T_{TCM}) should be adjusted lower so that the actual junction temperature (T_j) during the measurement is the same as that specified; thus, the measurement errors resulting from the junction temperature rise are corrected.

Step 3: Determining the initial reference current.

After the DUT is stabilized, the initial reference current, $I_F(0)$, is measured using one of the methods described in **Section 8.3**. If necessary, the measurement is repeated several times and the average initial reference current is used to reduce the uncertainty of the measured $I_F(0)$.

Note 4: After the initial reference current is measured, the DUT may be switched off for calculating $I_F(0)$ and/or pre-adjusting the TCM to an estimated temperature (lower than T_j), for operating the DUT at the specified T_j . A TCM is typically slow compared to the T_j rising speed; therefore,

the DUT may be operated at a T_j that is significantly higher than the specified value when the DUT is just turned on. If this overshooting of T_j is a concern, the TCM should be adjusted to an estimated temperature before the DUT is turned on, so that the DUT is operated at a T_j that is approximately equal to or lower than the specified T_j when the DUT is just turned on. The overshooting of T_j can also be avoided by turning the DUT on gradually (see **Note 5** in the next step).

Step 4: Operating the DUT at the specified T_j .

The DUT is turned on if it was switched off after Step 3. As the DUT continues to operate on the rated AC voltage, the subsequent reference current, $I_F(n)$, is measured using the same method used for measuring the initial reference current, $I_F(0)$. The measured subsequent reference current, $I_F(n)$, is maintained equal to the initial reference current, $I_F(0)$, by adjusting the TCM temperature, T_{TCM} , to hold the junction temperature constant at its initial value (see **Figure 2**).

Note 5: If the overshooting T_j is a concern, the DUT should be turned on gradually by increasing the duty cycle of the AC power from 0% to 100% so that T_j is always near the specified value.

Step 5: Performing optical and electrical measurements.

After the DUT is controlled at the specified junction temperature, optical and electrical measurements are performed for total luminous flux, total radiant flux, total photon flux, color characteristics, electrical voltage, current, and power, using conventional optical and electrical measurement instruments designed for measuring light sources operated under a normal AC operating condition. Because the first half of an AC cycle and the second half of the AC cycle may energize a different group of p-n junctions, optical and electrical measurements shall cover a period that is an integer number of AC cycles; e.g., 1 AC cycle period, 2 AC cycle periods.

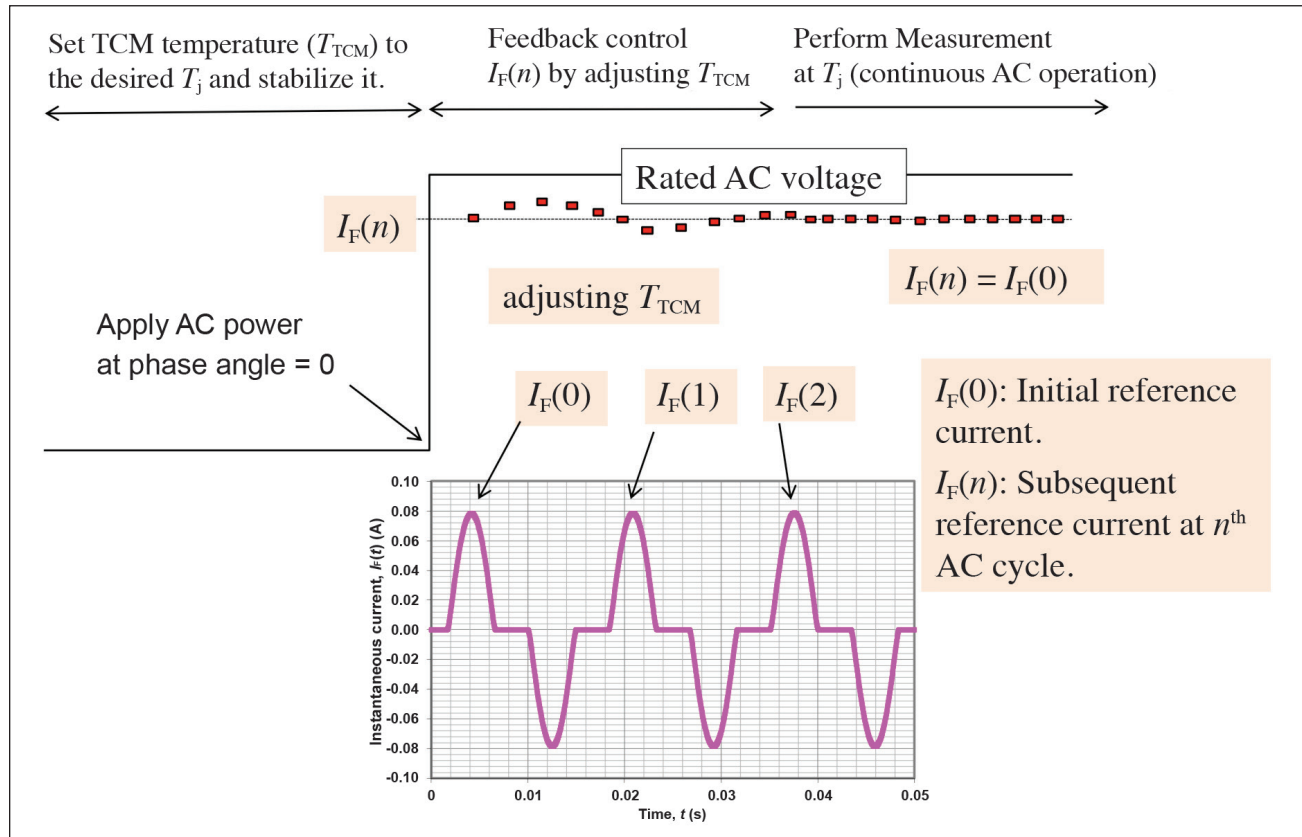


Figure 2. Illustration of the procedure to set a DUT at a specified T_j .

8.3 Methods Used to Determine the Initial Reference Current, $I_F(0)$

The initial reference current, $I_F(0)$, should be determined using one of the methods described below by applying the specified AC voltage at the 0 degrees phase angle to the DUT and measuring the instantaneous current, $i_F(t)$, and instantaneous voltage, $u_F(t)$.

8.3.1 RMS Initial Reference Current, $I_F(0)_{RMS}$. A simple way to test a DUT at a specified junction temperature, T_j , when using an AC power supply is to use the RMS current value of the first half of the first AC cycle, $I_F(0)_{RMS}$, as the initial reference current, $I_F(0)$, for feedback control. Instantaneous current, $i_F(t)$, over the first half of an AC cycle can be calculated/measured using a fast digital multimeter (e.g., 20,000 readings/s) and; thus, $I_F(0)_{RMS}$ can be calculated. Because the junction of the DUT is heated up during the first half of the AC cycle, the actual set junction temperature may be considerably higher than the specified T_j .

8.3.2 Instantaneous Initial Reference Current, $I_F(0)_{INS}$. The junction temperature of a DUT can be

set more accurately if the first instantaneous current $I_F(0)_{INS}$ at the “on” threshold voltage (e.g., 100 V) is measured accurately and used as the initial reference current, $I_F(0)$. At the threshold voltage of a DUT, the instantaneous current is small; thus, the self-heating of p-n junctions is insignificant, but a small increase in voltage will result in a large increase in current. In practice, it is difficult to measure the first instantaneous current, $I_F(0)_{INS}$, [and subsequent instantaneous currents, $I_F(n)_{INS}$] accurately because the instantaneous current is small and because both voltage and current change rapidly. To reduce the difficulty in measuring the instantaneous current, one of the two alternative methods described below can be used.

8.3.2.1 Using a Programmable AC Power Supply. Most programmable AC power supplies can generate a modified AC waveform, as shown in **Figure 3**. Using such an AC power supply greatly reduces the difficulty in measuring the initial and subsequent reference currents. The two short, flat lines (at the beginning of the first half of their AC power cycle) represent the

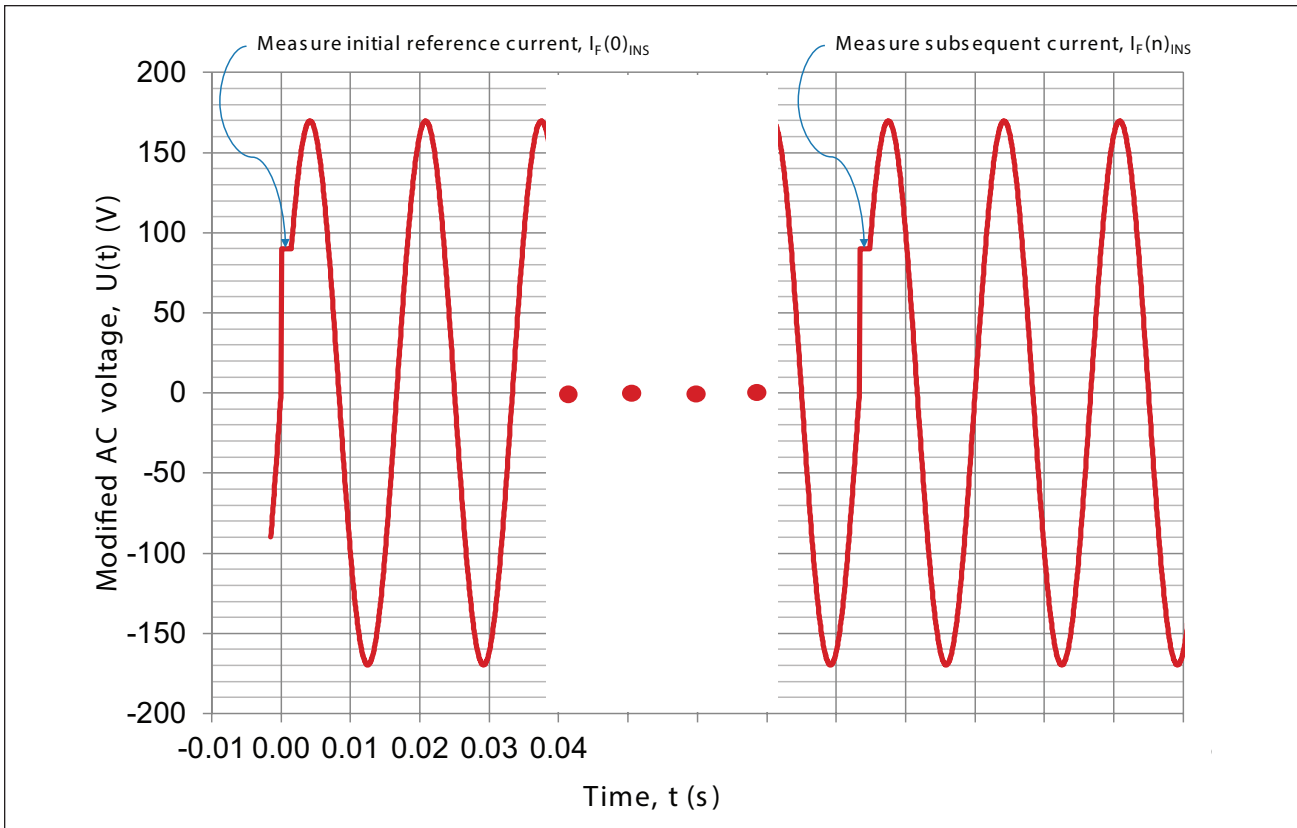


Figure 3. Modified AC waveform used for measurement of a DUT.

threshold voltage when the current through the DUT is still low compared to the peak current of the DUT (typically on the order of 0.1% of the peak current); thus, it does not introduce considerable self-heating and has negligible influence on the measured optical quantities, input current, and power values of the DUT. This low instantaneous current measured at the threshold voltage is the monitor current. The first monitor current, $I_F(0)_{\text{INS}}$, before the DUT is heated up, is used as the initial reference current, $I_F(0)$. While the DUT is operated on the AC power supply, the subsequent reference current at AC power cycle n , $I_F(n)=I_F(n)_{\text{INS}}$, is measured and used for feedback control by adjusting the TCM temperature, TTCM, so that $I_F(n)_{\text{INS}}$ is equal to $I_F(0)_{\text{INS}}$, which indicates that the temperature of the junction at AC power cycle n is the same as that before the DUT is turned on.

Note 6:

- a. *It is unnecessary to modify the voltage waveform and measure the subsequent reference current at every AC cycle because a temperature-controlled mount (TCM) is a slow device, and the junction temperature of a DUT changes slowly after it is stabilized; therefore, the number of modified voltage waveforms per second should be as small as possible—e.g., 1 modified voltage waveform per second for AC power of 60 Hz—so that the overall AC waveform is very similar to a normal AC waveform, which can further reduce the possible influence on the measured optical quantities, input current, and power values of the DUT.*
- b. *To cancel possible errors on the measured initial and subsequent reference currents resulting from the sudden voltage surge at the beginning of each AC cycle, the AC power may be applied to the DUT starting from a negative phase angle (a negative voltage) when the absolute AC voltage already falls below the DUT's threshold voltage, so that the voltage surge at the first AC cycle and those at subsequent AC cycles are exactly the same. Such an auxiliary voltage waveform before the phase angle 0 is shown as the dashed line in **Figure 3**.*

8.3.2.2 Using a Fast Source Measure Unit (SMU). When a fast SMU is used, the initial instantaneous reference current can be measured accurately without the need to modify the AC voltage waveform as described in **Section 8.3.2.1**. For example, an SMU was used to generate a digitized 60-Hz AC voltage waveform with 72

points per AC cycle—i.e., 4,320 samples per second—to drive a DUT and at the same time to measure the instantaneous current, $i_F(t)$, at each voltage level, $u_F(t)$, using a source-wait-measure scheme. The voltage and current AC waveforms are shown in **Figure 4**. Using such an SMU, the initial reference current, $I_F(0)$, can be determined based on the first instantaneous current, $I_F(0)_{\text{INS}}$, measured immediately after the voltage passes the DUT's "on" threshold voltage and before the DUT is heated up considerably.

$I_F(0)_{\text{INS}}$ may be obtained using a data processing technique such as averaging, fitting, or interpolating of the first several instantaneous currents, to reduce the influence of measurement noise. In addition, for producing a reasonably smooth AC waveform, the SMU's speed should be increased to at least 167 points per 60-Hz AC cycle (200 points for 50-Hz AC cycle), equivalent to a speed of 10,000 sources and measures per second; in addition, the voltage, current, and optical-signal waveforms should be checked using an oscilloscope to ensure no spikes occur during the point-to-point transient time.

8.4 Estimation of the Measurement Uncertainty Due to Junction Temperature Rise

When the RMS current over the first half of the first AC cycle, $I_F(0)_{\text{RMS}}$ (see **Section 8.3.1**), is used as the initial reference current, $I_F(0)$, the rise of the junction temperature, T_j , over the first half of the AC cycle leads to a higher than specified junction temperature when the DUT reaches its thermal equilibrium. Like the estimation of uncertainty for the single-cycle AC operation (**Section 7.3**), the difference between the specified T_j and the actual T_j should be estimated using a proper thermal model (e.g., using the model described in **Informative Reference 7**) if all design details and parameters of the DUT are known or can be measured using a proper method (e.g., using the method described in **Informative Reference 8**). Using the estimated difference, the errors resulting from the junction temperature rise can be corrected during the measurement time (see **Section 8.2, Step 2**). Otherwise, the measurement errors related to the junction temperature difference should be analyzed using the sensitivity coefficients of the junction temperature of the DT for optical and electrical quantities. Such

measurement errors can also be estimated by comparing the measurement results with those obtained using the instantaneous initial reference current, $I_F(0_{\text{INS}})$ (see **Section 8.3.2**), which has no, or negligible, errors resulting from the rise of junction temperature during the first half of the first AC cycle.

Corrections should be applied in order to reduce measurement uncertainties. When a correction is applied, the correction method and correction factor shall be reported, and the estimated residual error should be listed as an uncertainty component contributing to the overall measurement uncertainty. If the correction is not applied, this measurement error should be considered as a component of the overall measurement uncertainty.

When the first instantaneous current, $I_F(0_{\text{INS}})$ (see **Section 8.3.2.1** and **Section 8.3.2.2**), is used for determining the initial reference current, $I_F(0)$, the rapid rise of the junction temperature over the first half of the first AC cycle has no, or negligible, influence on setting the junction temperature of the DUT; therefore, the correction for measurement error due to the junction temperature rise is not needed.

9.0 Optical Measurements

The methods used for optical measurements described in IES LM-85-14 (see **Section 2.1**) shall be used for measurement of total luminous flux, total radiant flux, total photon flux, and color characteristics. The measured optical values shall be time-averaged values.

10.0 Test Report

The report shall include the information listed below from **Section 10.1** through **Section 10.8**. Total spectral radiant flux (unit: W/nm)* shall also be reported if

* Total spectral radiant flux: The spectral radiant flux emitted from a source and integrated over all directions (4π sr).

measured, and can be used to calculate all optical quantities listed in **Section 10.7**. The uncertainties of optical and electrical measurement results shall be reported if requested.

10.1 General Information

- Name of test laboratory
- Test date
- Report issue date (optional)

10.2 Description of DUT

- Designation and/or serial number
- Manufacturer (if known)
- Model number (if known)
- Type of DUT (package, array, or module, or remote-phosphor type DUT) (if known)

10.3 Description of Measurement System

- Optical measurement instruments
- Electrical measurement instruments
- Reference standard(s) for the measurement system

10.4 Environmental Conditions

- Ambient air temperature
- Room humidity (recommended to be reported)
- Ambient airflow speed (measured or estimated) if a remote-phosphor type DUT (recommended to be reported)

10.5 Measurement Conditions

- Operating conditions and durations if DUT is seasoned before the measurement
- Operating orientation (if a remote-phosphor type DUT)
- DUT junction temperature

10.6 Description of Measurement

- Test method (the single-cycle AC operation [**Section 7**], or continuous-cycle AC operation [**Section 8**])
- Initial reference current (RMS initial reference current [**Section 8.3.1**], or instantaneous initial reference current [**Section 8.3.2**]), if the continuous-cycle AC operation method is used

- The threshold voltage, monitor current, and number of the modified voltage waveforms per second, if the modified AC waveform [Section 8.3.2.1] is used
- Measurement procedure
- Applied corrections on the measurement results for errors such as self-absorption error when an integrating sphere is used, spectral mismatch error when a photometer is used, and those due to the junction temperature rise and “cold” phosphor temperature (if applicable) (see Section 7.3 and Section 8.4)

10.7 Optical Quantities (Time-Averaged) (Typical)

- Total luminous flux (lm)
- Total radiant flux (W)
- Total photon flux (mol/s)
- Luminous efficacy (lm/W)
- Chromaticity coordinates
- Correlated color temperature (CCT), if a white DUT
- $D_{uv}^{\dagger}$, if a white DUT
- Color rendering index (CRI), if a white DUT
- Dominant wavelength, if a single-color DUT
- Centroid wavelength, if a single-color DUT
- Peak wavelength, if a single-color DUT

10.8 Electrical Quantities (Typical)

- Input RMS voltage (V)
- Input RMS current (A)
- Input real power (W)
- Frequency of the AC power

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[†] “ D_{uv} ” (symbol: D_{uv}) is defined as the closest distance from the Planckian locus on the (u' , $2/3v'$) diagram, with “+” for above and “-” for below the Planckian locus.⁹

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